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1 C:\Users\270708326\PycharmProjects\PythonProject\SmartHealthCare\.venv\Scripts\python.exe C:\
  Users\270708326\PycharmProjects\PythonProject\SmartHealthCare\data_preprocessing_cleaning.py
2 =====
3 SMARTCARE DATA PREPROCESSING & CLEANING
4 =====
5 Dataset loaded successfully
6 Records: 50
7 Variables: 13
8
9 =====
10 DATA QUALITY ASSESSMENT
11 =====
12
13 First 5 records
14 PatientID  Age  Gender  ...  FallAlerts  CalorieBurn  WeeklyHealthScore
15 0      P1000   56  Female  ...           1           3282             99
16 1      P1001   69   Male  ...           0           1757             94
17 2      P1002   78  Female  ...           0           1996             60
18 3      P1003   64  Female  ...           0           2395             78
19 4      P1004   60   Male  ...           0           1659             85
20
21 [5 rows x 13 columns]
22
23 Data Types:
24 PatientID                object
25 Age                      int64
26 Gender                   object
27 DailyStepCount           int64
28 HeartRate_Min            int64
29 HeartRate_Max            int64
30 HeartRate_Avg            int64
31 SleepDuration_Hours      float64
32 SleepQualityScore        int64
33 StressIndex              int64
34 FallAlerts               int64
35 CalorieBurn               int64
36 WeeklyHealthScore        int64
37 dtype: object
38
39 Descriptive Statistics:
40      Age  DailyStepCount  ...  CalorieBurn  WeeklyHealthScore
41 count  50.000000      50.000000  ...    50.000000             50.00000
42 mean   67.160000      7482.260000  ...   2569.720000             77.88000
43 std     8.970155      2830.756644  ...    606.615503             13.12054
44 min    50.000000      2064.000000  ...   1511.000000             56.00000
45 25%    60.000000      5100.250000  ...   2107.000000             68.00000
46 50%    69.500000      7783.500000  ...   2543.500000             77.50000
47 75%    75.000000      9910.000000  ...   3118.750000             86.00000
48 max    79.000000     11762.000000  ...   3452.000000             99.00000
49
50 [8 rows x 11 columns]
51
52 Missing Values Check:
53 PatientID                0
54 Age                      0
55 Gender                   0
56 DailyStepCount           0
57 HeartRate_Min            0
58 HeartRate_Max            0
59 HeartRate_Avg            0
60 SleepDuration_Hours      0
61 SleepQualityScore        0
62 StressIndex              0
63 FallAlerts               0
64 CalorieBurn               0
65 WeeklyHealthScore        0
66 dtype: int64
67 No missing values detected
68
```

```
69 Duplicate Records: 0
70 No duplicate records found
71
72 =====
73 FEATURE ENGINEERING
74 =====
75
76 Age groups created
77 AgeGroup
78 50-59      11
79 60-69      14
80 70-79      25
81 Name: count, dtype: int64
82
83 Activity levels created
84 ActivityLevel
85 Moderate    26
86 Sedentary   12
87 Active      12
88 Name: count, dtype: int64
89
90 Sleep categories created
91 SleepCategory
92 Good        23
93 Poor         14
94 Fair         13
95 Name: count, dtype: int64
96
97 Heart Rate Range calculated
98 Mean HR Range: 74.62 bpm
99
100 Stress levels created:
101 StressLevel
102 High         21
103 Moderate     17
104 Low          12
105 Name: count, dtype: int64
106
107 Health categories created:
108 HealthCategory
109 Fair         20
110 Good         15
111 Excellent    11
112 Poor          4
113 Name: count, dtype: int64
114
115 Composite Risk Score created
116 Mean Risk Score: 52.42
117 Risk Score Range: 0.00 - 100.00
118
119 ✓ Risk categories created:
120 RiskCategory
121 High Risk     19
122 Medium Risk   18
123 Critical Risk  7
124 Low Risk      6
125 Name: count, dtype: int64
126
127 =====
128 OUTLIER DETECTION (IQR Method)
129 =====
130
131 ✓ No significant outliers detected
132
133 =====
134 DATA VALIDATION
135 =====
136
137 Validation Results:
```

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138     Age Range (50-100): ✓ PASS
139     HR Min < HR Max: ✓ PASS
140     Sleep Duration (0-12h): ✓ PASS
141     Stress Index (0-100): ✓ PASS
142     Fall Alerts (0/1): ✓ PASS
143     Health Score (0-100): ✓ PASS
144
145 ✓ All validation checks passed
146
147 =====
148 DATA EXPORT
149 =====
150
151 ✓ Cleaned dataset saved: 'smartcare_cleaned_data.csv'
152
153 Data Summary:
154     Total Records: 50
155     Total Variables: 21
156     Original Variables: 13
157     Engineered Features: 8
158     Missing Values: 0
159     Duplicates: 0
160     Age Range: 50-79
161     Gender Distribution: {'Female': 30, 'Male': 20}
162     Fall Alerts: 5
163
164 =====
165 PREPROCESSING COMPLETE
166 =====
167
168 Next Steps:
169     1. Run exploratory data analysis (EDA)
170     2. Perform correlation analysis
171     3. Conduct statistical testing
172     4. Build predictive models
173 =====
174
175 Process finished with exit code 0
176
```